

SCHOOL OF COMPUTER & SYSTEMS SCIENCE

MCA - Sem III

CS- 414: Theoretical Computer Science Mid-Semester Examination

Date: 28-10-2023

Time: 10:00am – 12:00noon

Max. Marks: 25

Part-A [15]

ALL QUESTIONS ARE COMPULSORY. Questions 1 – 5 carry 1 mark each, and questions 6 – 10 carry 2 marks each.

- 1) Give a **wff** using the propositional variables **P, Q** and **R**.
- 2) State the **Absorption Law** in set theory.
- 3) Consider the function **f: X → Y** such that **X = {a, b, c, d}** and **Y = {1, 2, 3}**.
Can **f** be *one-to-one*, *onto* or both? Justify your answer.
- 4) In the production rule say, $\alpha \rightarrow \beta$ in $G = (N, \Sigma, P, S)$, why is $\alpha \notin \Sigma$?
- 5) Prove the set theoretic inequality $|S \cup T| \leq |S| + |T|$ by an illustration.
- 6) Give the **Conjunctive Normal Form** or **Disjunctive Normal Form** of the formula $((A \wedge B) \rightarrow C) \wedge D$.
- 7) For disjoint sets **A** and **B**, prove the *closure property* $(A^* \cup B^*)^* = (A \cup B)^*$.
- 8) Is the given statement – “*The set inclusion ‘ $\subseteq$ ’, on the power set of a set S is a total ordering*” true or false? Justify.
- 9) Given a transition function, $\delta: Q \times \Sigma \rightarrow Q$ of a DFA $M = (Q, \Sigma, \delta, q_0, F)$, describe $\hat{\delta}(w, q_i)$ or $\vdash (w, q_i)$.
- 10) Consider a Vending Machine that can dispensing two products **Coffee** and **Tea**. Let price of both be unique, say **c** and **d**. What kind of Finite State Machine is this Vending Machine? Justify your answer.

Part-B [10]

ANSWER ANY TWO OUT OF THE FOLLOWING. Each carry 5 marks.

- 11) a) Consider an alphabet $\Sigma = \{0, 1\}$. Consider the regular expressions $r_1=01$, $r_2=001$, $r_3=010$, and $r = (r_1 + r_2 + r_3)^*$. Give the transition diagrams T_1 , T_2 , T_3 and T to accept all strings in $L(r_1)$, $L(r_2)$, $L(r_3)$ and $L(r)$ respectively.
- b) Determine if the finite automaton M equivalent to T is deterministic or nondeterministic. [4+1]
- 12) a) In the Pumping lemma for Regular languages, say L , there exists an integer $n \geq 1$ such that for a $w \in L$, and $\|w\| \geq n$ and $w = xyz$. What are the other conditions, and what is the value of n ? Prove that all $xy^iz \in L$.
- b) Give one strategy used in minimization of finite state machines. [3+2]
- 13) a) Consider $L(G)$ a language of strings with equal number of a 's and b 's. Let the $G = (N, \Sigma, P, S)$ be the context-free grammar, where $N = \{S, A, B\}$, $\Sigma = \{a, b\}$. Give the additional production rules to the following:
- $S \rightarrow aB$
 $S \rightarrow bA$
:
- b) Derive the string **bbabaaba**. [3+2]

School of Computer & Systems Science
MCA - Sem III
CS- 414: Theoretical Computer Science
End-Semester Examination

Date: 14-12-2023

Time: 10:00 – 01:00pm

Max. Marks: 50

PART – A [20]

ALL questions are compulsory. Each carry 2 marks.

1. Consider a regular set defined over $\Sigma = \{a, b\}$. Give the transition graphs equivalent to the following regular expressions:

- a. $r_1 = a$
- b. $r_2 = b + a$
- c. $r_3 = ab$
- d. $r_4 = b^*$

2. Write a grammar for the language $L_1 = \{w^Rcw \mid w \in (0+1)^*\}$.

3. Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

4. Consider two sets $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$, the two subsets of the universal set U . If the bit string to represent A is 11 1110 0000 then give the bit strings to represent U and B . Using suitable operations on bit strings, prove that element 3 $\in A \cap B$.

5. Determine if a binary relation R , defined on $N = \{1, 2, 3, \dots\}$, the set of natural numbers, is an equivalence relation,

$$R = \{(x, y) \mid x, y \in N \text{ and } x + y \text{ is even}\}$$

If ^{*R is*} an equivalence relation, give the equivalence classes?

6. Design a Moore machine M over an input alphabet $\{a, b\}$ with $\lambda: Q \rightarrow \{0,1\}$ to count the number of occurrences of the substring aab, so that, (18) ^{*90*} 000010010000010000 must be the output for the input baabaabbabaabbaba. (17)

You may use the transition graph to present the transition functions δ and λ .

7. What are the various types of functions? Is **predicate** a function? If Yes, then specify its type. If No, then justify your answer.

8. Consider the grammar $G = (\{E\}, \{a, (,)\}, P, E)$, comprising the production rules $P: E \rightarrow E + E \mid E * E \mid (E) \mid a$. G is an ambiguous grammar therefore give ^{*unambiguous*} G' , a grammar equivalent to G such that $L(G) = L(G')$.

9. Define the transition function of a Deterministic Push Down Automata. Also give the transition function δ for, ^{*D P D A*} $Q \times \Sigma \times \Gamma \rightarrow Q \times \mathbb{F}^*$

- a. push **A** to the stack, on reading **a** on the input tape with **X** the top symbol
 b. pop **B** from the stack, on reading **b** on the input tape.
10. Define Chomsky Normal Form. Give its significance in study of formal languages. *CFG to CNF. Remove ambiguity. → Remove null prodn. remove unit*

PART - B [30]

Answer any **THREE** of the following. Each carry 10 marks.

11. a. Give a Context Free Grammar G to generate all strings of the language L_2 over $\{0, 1\}$. L_2 contains equal numbers of 1's and 0's. Give the derivation of one valid string.
 b. For any two integer values m and n give a Turing Machine ^{over $\{0, 1\}$} that may compute $m > n$. *push and then pop* [5+5]
12. a. Describe how a **Top-Down Parser** operates.
 b. State one closure property of CFLs and prove using one of these options – using **Pumping Lemma for CFL**, or **designing a PDA**, or **designing a CFG**? [4+6]
13. a. State the Pumping Lemma for Finite Automata
 b. Let $L_3 = \{a(ab + ba)^*b\}$ $\Sigma = \{a, b\}$. Design a Finite Automata to recognize the strings of the set $(\Sigma^* \setminus L_3)$ (complement of L_3).
 c. Determine if $(\Sigma^* \setminus L_3)$ is a Regular Set. [3+4+3]
14. a. Define an **Alphabet**, **Production rule of a Context Sensitive Grammar**, and a **Recursively Enumerable Set**.
 b. What is the language L_4 accepted by the given by Turing Machine of which the transition functions given below?
 c. Also determine if L_4 is Recursive set or a Recursively Enumerable Set. [3+4+3]

